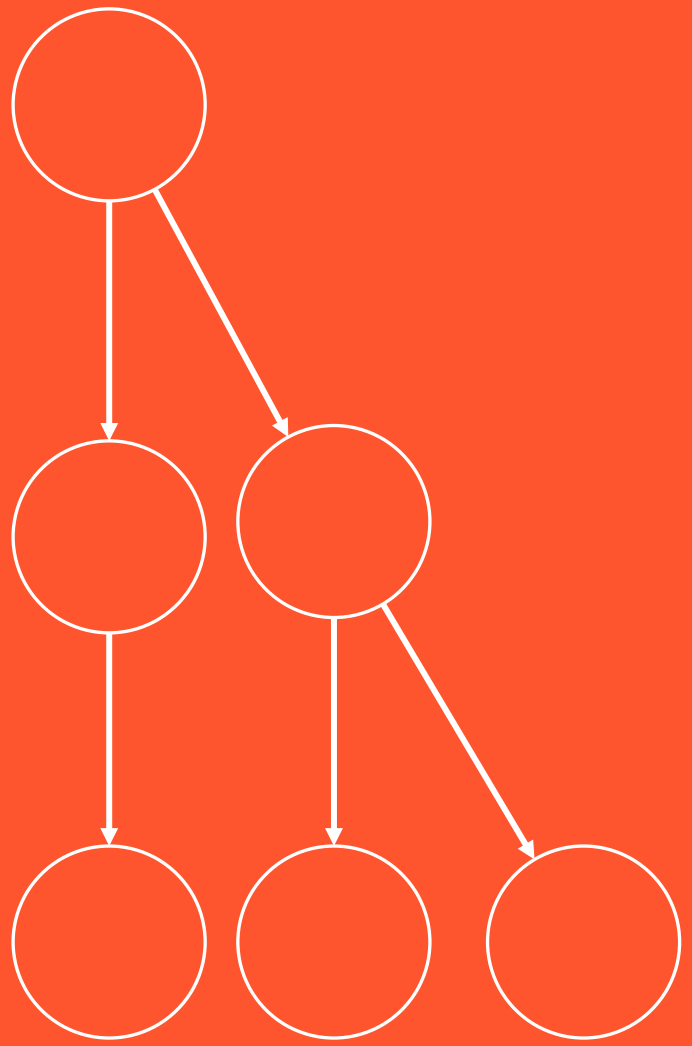


BCOG 100

Pattern Recognition

Lecture 7.1 — The Pattern Recognition Problem

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

To Start: Never the Same Twice

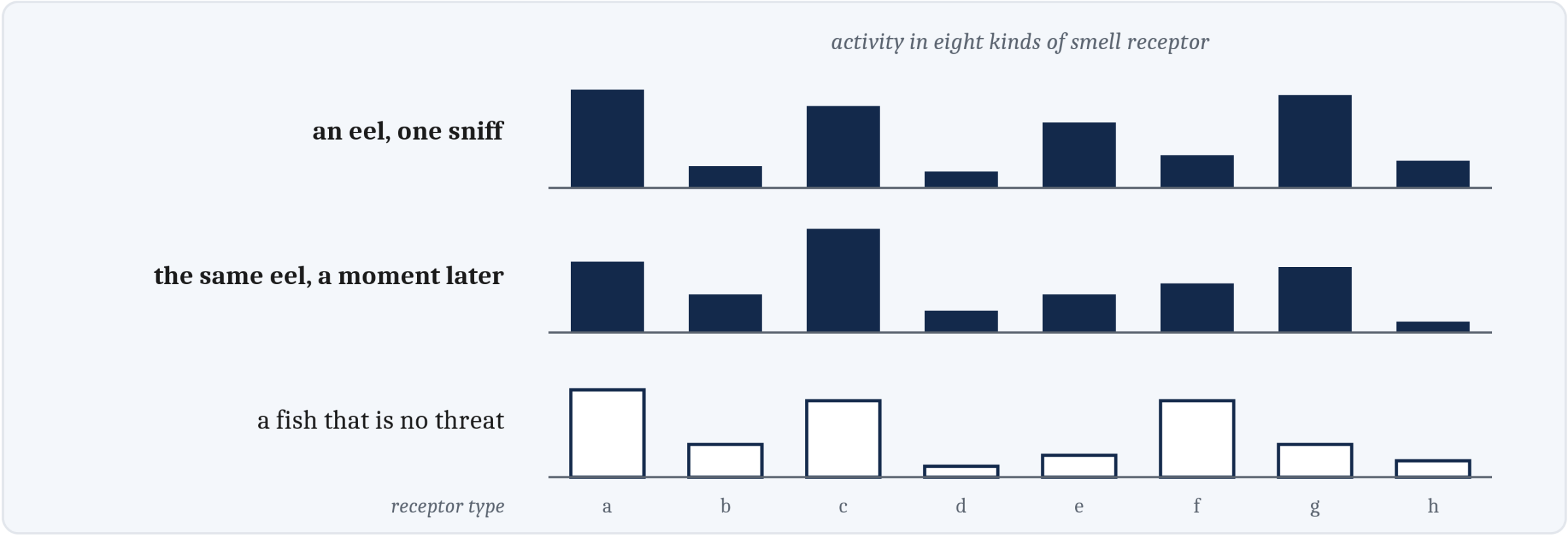
In your group, three minutes.

What is a smell you always recognize? No two encounters with it are ever exactly the same.

What stays the same, so that it can still be recognized? And what nearly identical thing would have to be told apart from it?

No Detector for an Eel

An eel's smell is a pattern across many receptors, never twice the same, and much of it shared with harmless fish.

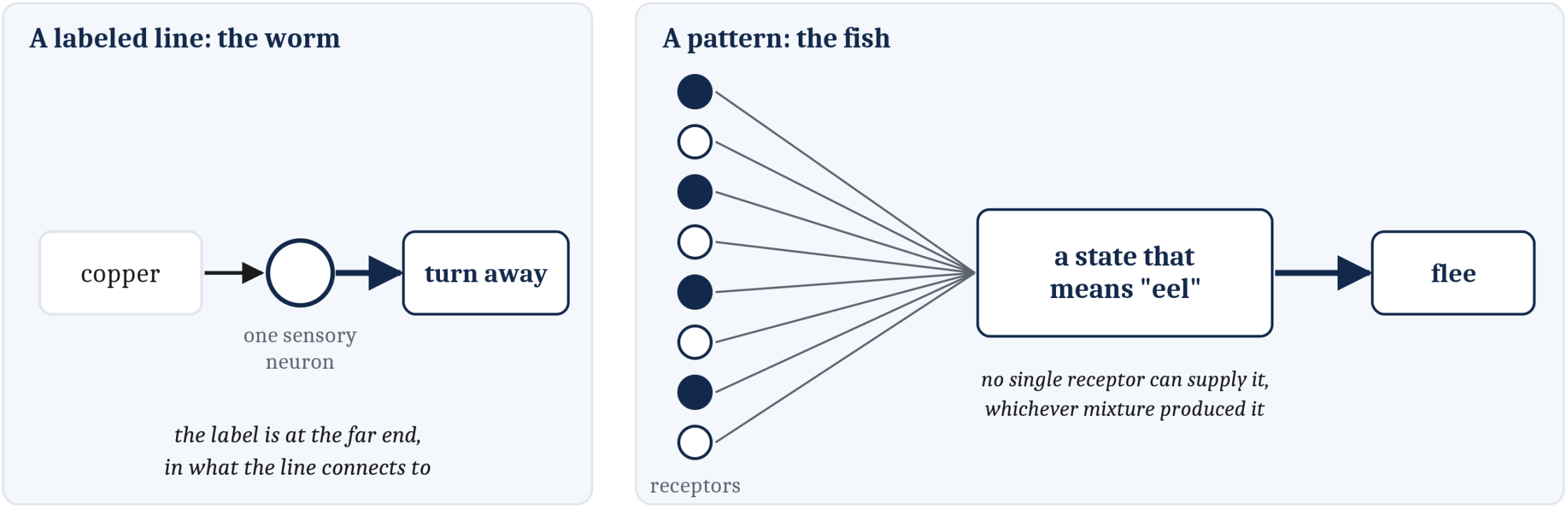


No receptor type is the eel receptor. Every one of them also responds to the harmless fish.

Invented numbers: the patterns show a shape of argument, not a measurement.

Two Ways to Recognize Something

Pattern recognition needs something between receptors and muscles: a **representation**, activity that stands for the eel whichever mixture produced it.



Representation: a pattern of activity that stands for something in the world,
so that what the animal does next depends on what the pattern stands for.

Telling Apart, Treating Alike

Discrimination: inputs calling for different responses must be told apart. **Generalization**: inputs calling for the same response must be treated alike.

Generalization: treat alike

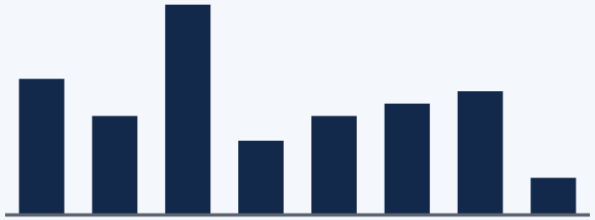
different mixtures, the same response

a large eel, upstream



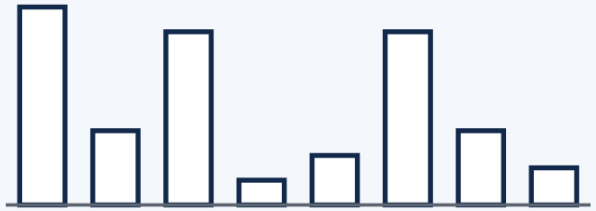
flee

a small eel, downstream



flee

a sucker: harmless



keep feeding

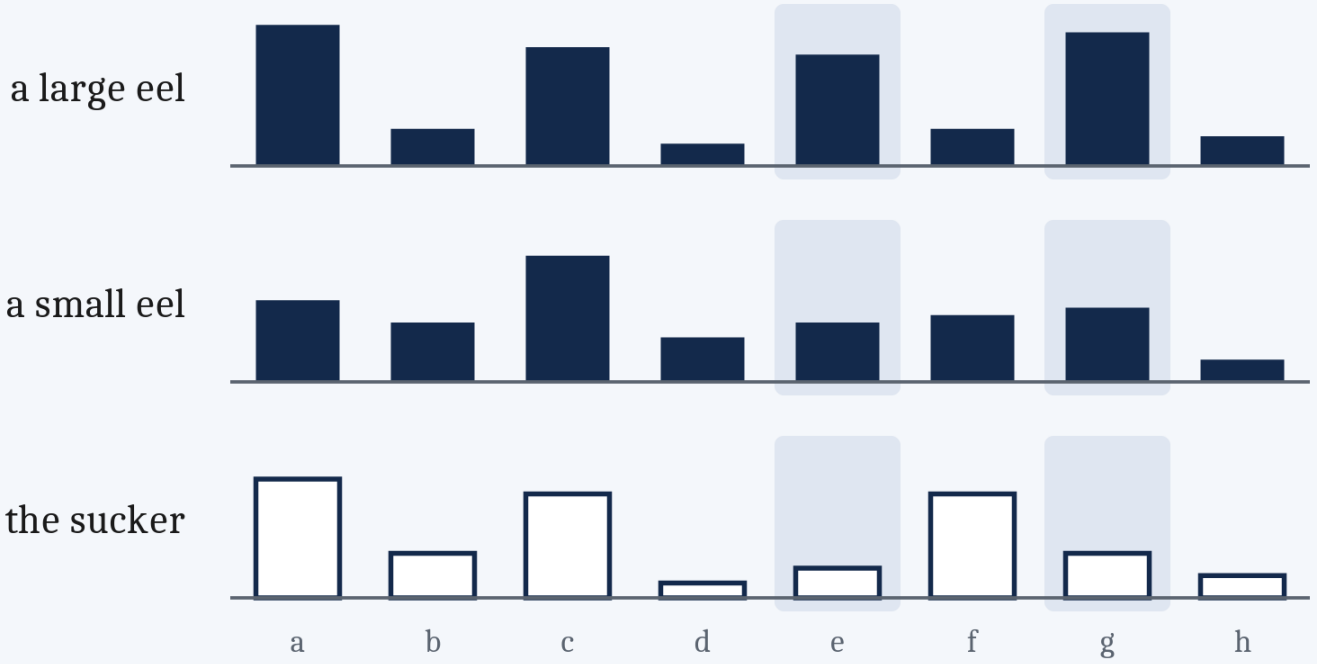
Discrimination: tell apart

much the same chemistry, different responses

Filled bars: an eel. Hollow bars: the sucker.

Two Demands, One Mapping

The two demands push on the same inputs in opposite directions. Some differences have to come to count for more, and others for less.



The shaded receptor types are where eel and sucker differ most.

Sharpen on them,
and the sucker comes apart from the eel, but the large and small eel differ there too.

Widen what counts as eel,
and every eel is caught, and so is the sucker.

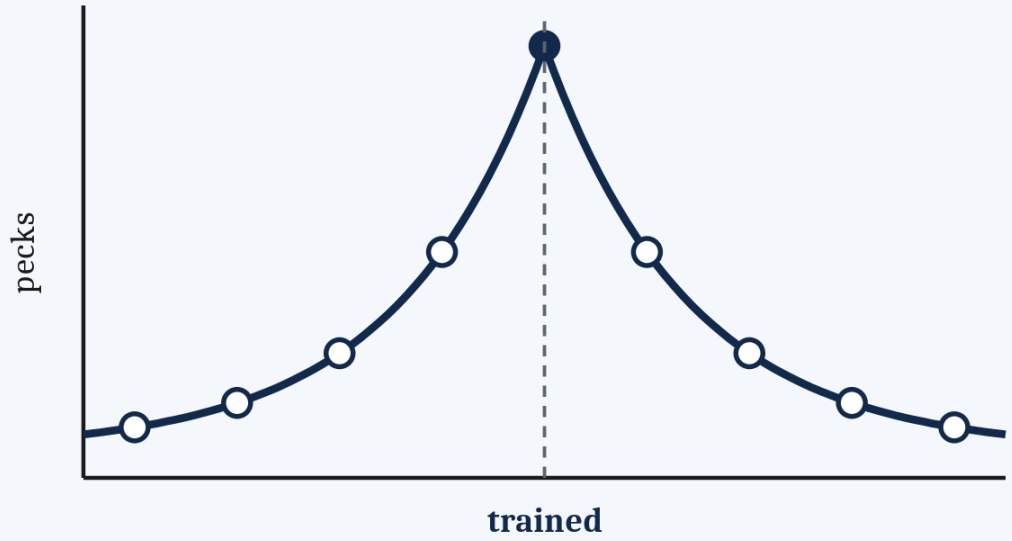
Neither more discrimination nor more generalization across the board will do.

Some differences have to come to count for more, and others for less.

How Alike Does Something Have to Be?

A **generalization gradient** is a description of behavior: how strongly the trained response is given, as the test input differs more from the trained one.

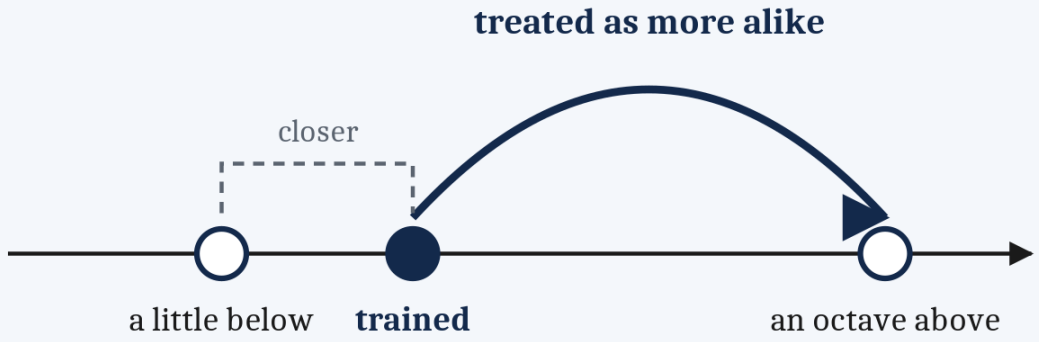
Measured as behavior



how far the test color is from the trained one

schematic: the shape, not real data

Closeness in physical units can mislead



pitch, in physical units

Which differences matter has to be measured by watching what the animal does.

A generalization gradient describes what the animal does. Nothing in it is about what happens inside.

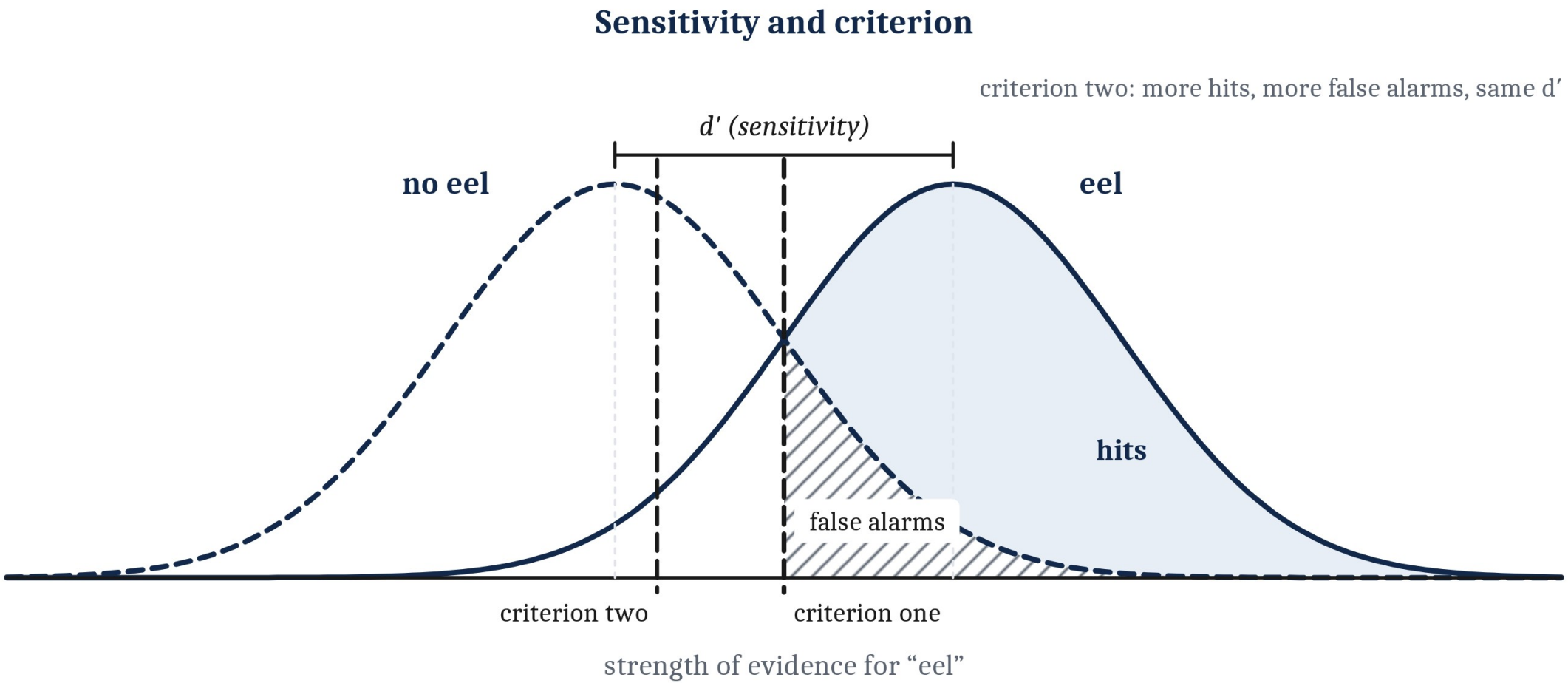
Deciding Under Noise: Four Outcomes

Signal detection theory: every detection decision is a **hit**, a **miss**, a **false alarm**, or a **correct rejection**.

	The fish flees	**The fish stays**
An eel is present	Hit	Miss
No eel is present	False alarm	Correct rejection

Two Quantities, Not One

Sensitivity: how far apart the two distributions lie (*d′*). **Criterion**: where the line is drawn. Moving the criterion never changes sensitivity.



Where Should the Criterion Go?

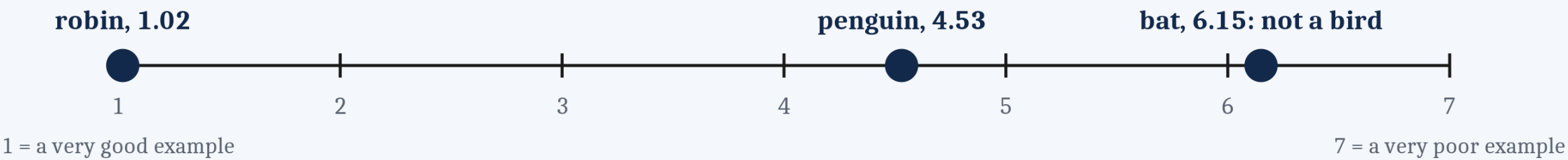
It depends on what each outcome gains or loses, and on how common targets are. Accuracy is the wrong measure when the two errors cost different amounts.

Case	What each error costs	Best criterion
The small fish	A miss may end its life; a false alarm loses a few minutes of feeding	Liberal
Rare targets	False alarms lose a great deal	Conservative
A worked example	A false alarm loses 15 points; a miss loses 1	Conservative: accuracy peaks at neutral, but points peak here
A dim room	Sensitivity is poorer	More extreme in the same direction

Categories Have Structure

A **category** is a set of things treated as the same kind, for some purpose. **Typicality**: some members are better examples than others.

How good an example of a bird?



Family resemblance

One member shares some features with a second, the second shares others with a third, and nothing runs through the whole set.

The most typical share most with their own kind.

Not held together by resemblance

Built around a goal: things to take out of a burning house.

Defined by a relation: a predator, by what it does to other animals.

Ratings from 209 students; different groups agreed closely.

When Only Combinations Matter

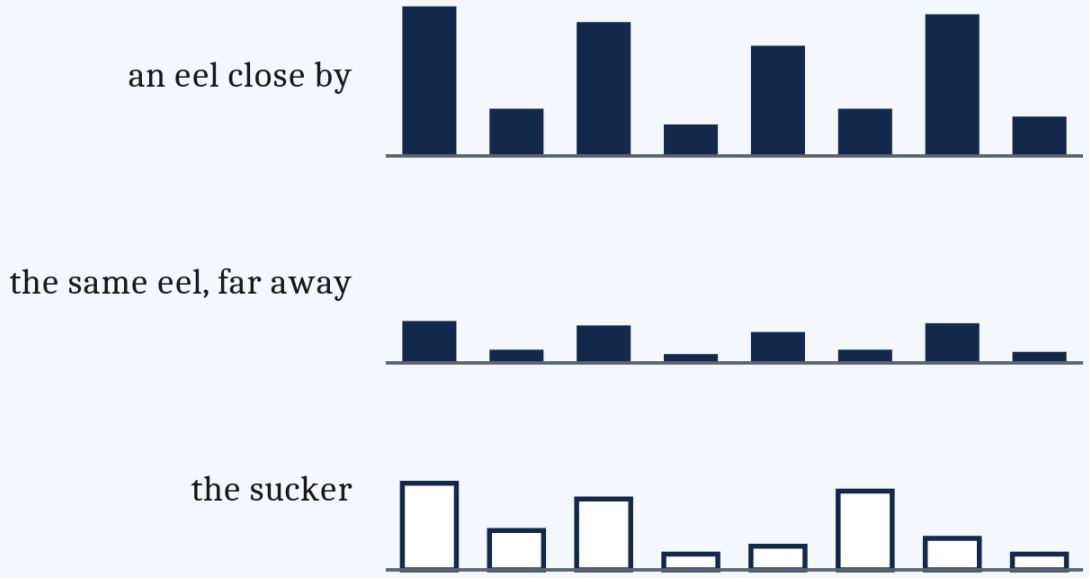
For some categories, no single input carries any information. The fruit-eater's rule is *exclusive or* (XOR): one smell or the other, but not both.

	Smell B absent	**Smell B present**
Smell A absent	No fruit	Ripe fruit
Smell A present	Ripe fruit	Rotting fruit

The Hardest Cases: Invariance, and Learning in Sequence

Invariance: recognizing one thing across changes that leave its inputs barely overlapping. And new learning has to add to old without overwriting it.

One thing, inputs that barely overlap



The two eel smells can differ from each other
more than either differs from the sucker's.

Invariance: one neuron downstream fires for both.

Learning in sequence



Shaded: receptor types the two smells share.

Learning the new predator must not
erase what the fish knows about eels,
though both use many of the same receptors.

Module 5: stability versus plasticity

Invented numbers: the patterns show a shape of argument, not a measurement.

A Detector for AI-Written Essays

In your group, four minutes.

A company sells a detector that flags student essays as AI-written.

What are its four outcomes? Which error costs more, and to whom?

The company announces a new version that is “more sensitive”: it catches more AI-written essays. What else would we need to know before believing that its sensitivity improved?

Image Sources

Slide	Image	File	License	Source
3	No receptor stands for an eel	fig-never-twice.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
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